

INTRODUCTION A boundary-value problem is said to be **nonhomogeneous** if either the partial differential equation or the boundary conditions are nonhomogeneous. The method of separation of variables employed in the preceding three sections may not be applicable to a nonhomogeneous boundary-value problem *directly*. In the first of the two techniques examined in this section we employ a change of dependent variable $u = v + \psi$ that transforms a nonhomogeneous boundary-value problem into two BVPs: one involving an ODE and the other involving a PDE. The latter problem is homogeneous and solvable by separation of variables. The second technique may also start with a change of a dependent variable, but is basically a frontal attack on the BVP using orthogonal series expansions.

The two solution methods that follow are distinguished by different types of nonhomogeneous boundary-value problems.

Time Independent PDE and BCs We first consider a BVP involving a *time-independent nonhomogeneous equation* and *time-independent boundary conditions*. An example of such a problem is

$$\begin{aligned} k \frac{\partial^2 u}{\partial x^2} + F(x) &= \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0 \\ u(0, t) &= u_0, \quad u(L, t) = u_1, \quad t > 0 \\ u(x, 0) &= f(x), \quad 0 < x < L, \end{aligned} \quad (1)$$

where $k > 0$ is a constant. We can interpret (1) as a model for the temperature distribution $u(x, t)$ within a rod of length L where heat is being generated internally throughout the rod by either electrical or chemical means at a rate $F(x)$ and the boundaries $x = 0$ and $x = L$ are held at constant temperatures u_0 and u_1 , respectively. When heat is generated at a constant rate r within the rod, the heat equation in (1) takes on the form

$$k \frac{\partial^2 u}{\partial x^2} + r = \frac{\partial u}{\partial t}. \quad (2)$$

Now equation (2) is readily shown not to be separable. On the other hand, suppose we wish to solve the usual homogeneous heat equation $ku_{xx} = u_t$ when the boundaries $x = 0$ and $x = L$ are held at, say, nonzero temperatures. Even though the substitution $u(x, t) = X(x)T(t)$ separates the PDE, we quickly find ourselves at an impasse in determining eigenvalues and eigenfunctions because no conclusion about the values of $X(0)$ and $X(L)$ can be drawn from $u(x, 0) = X(0)T(t) = u_0$ and $u(x, L) = X(L)T(t) = u_1$.

By changing the dependent variable u to a new dependent variable v by the substitution $u(x, t) = v(x, t) + \psi(x)$, (1) can be reduced to two problems:

$$\text{Problem A: } \{k\psi'' + F(x) = 0, \quad \psi(0) = u_0, \quad \psi(L) = u_1\}$$

$$\text{Problem B: } \begin{cases} k \frac{\partial^2 v}{\partial x^2} = \frac{\partial v}{\partial t}, \\ v(0, t) = 0, \quad v(L, t) = 0 \\ v(x, 0) = f(x) - \psi(x) \end{cases}$$

Observe in Problem A that the simple ODE $k\psi'' + F(x) = 0$ can be solved by integration. Moreover, Problem B is a homogeneous BVP that can be solved straight away by the method of separation of variables. A solution of the given nonhomogeneous problem is then a superposition of solutions:

$$\text{Solution } u = \text{Solution } \psi \text{ of Problem A} + \text{Solution } v \text{ of Problem B.}$$

There is nothing in the above discussion that should be memorized; you should work through the substitution $u(x, t) = v(x, t) + \psi(x)$ each time as outlined in the next example.

$$\begin{aligned}
\text{Solve} \quad & k \frac{\partial^2 u}{\partial x^2} + r = \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\
& u(0, t) = 0, \quad u(1, t) = u_1, \quad t > 0 \\
& u(x, 0) = f(x), \quad 0 < x < 1,
\end{aligned} \tag{3}$$

where r and u_1 are nonzero constants.

SOLUTION Both the partial differential equation and the boundary condition at $x = 1$ are nonhomogeneous. If we let $u(x, t) = v(x, t) + \psi(x)$, then

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x^2} + \psi''(x) \quad \text{and} \quad \frac{\partial u}{\partial t} = \frac{\partial v}{\partial t}.$$

After substituting these results, the PDE in (3) then becomes

$$k \frac{\partial^2 v}{\partial x^2} + k\psi''(x) + r = \frac{\partial v}{\partial t}. \tag{4}$$

Equation (4) reduces to a homogeneous equation if we demand that ψ satisfy

$$k\psi''(x) + r = 0 \quad \text{or} \quad \psi''(x) = -\frac{r}{k}.$$

Integrating the last equation twice reveals that

$$\psi(x) = -\frac{r}{2k}x^2 + c_1x + c_2. \tag{5}$$

Furthermore,

$$u(0, t) = v(0, t) + \psi(0) = 0$$

$$u(1, t) = v(1, t) + \psi(1) = u_1.$$

We have $v(0, t) = 0$ and $v(1, t) = 0$ provided that ψ also satisfies

$$\psi(0) = 0 \quad \text{and} \quad \psi(1) = u_1.$$

By applying the latter two conditions to (5) we obtain, in turn, $c_2 = 0$ and $c_1 = r/2k + u_1$. Consequently

$$\psi(x) = -\frac{r}{2k}x^2 + \left(\frac{r}{2k} + u_1\right)x.$$

Finally, the initial condition in (3) implies $v(x, 0) = u(x, 0) - \psi(x) = f(x) - \psi(x)$.

Thus to determine $v(x, t)$ we solve the new boundary-value problem

$$\begin{aligned}
& k \frac{\partial^2 v}{\partial x^2} = \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\
& v(0, t) = 0, \quad v(1, t) = 0, \quad t > 0 \\
& v(x, 0) = f(x) + \frac{r}{2k}x^2 - \left(\frac{r}{2k} + u_1\right)x, \quad 0 < x < 1
\end{aligned}$$

by separation of variables. In the usual manner we find

$$v(x, t) = \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x, \tag{6}$$

$$\text{where} \quad A_n = 2 \int_0^1 \left[f(x) + \frac{r}{2k}x^2 - \left(\frac{r}{2k} + u_1\right)x \right] \sin n\pi x \, dx. \tag{7}$$

A solution of the original problem is $u(x, t) = v(x, t) + \psi(x)$, or

$$u(x, t) = -\frac{r}{2k}x^2 + \left(\frac{r}{2k} + u_1\right)x + \sum_{n=1}^{\infty} A_n e^{-kn^2\pi^2 t} \sin n\pi x, \quad (8)$$

where the coefficients A_n are defined in (7). ≡

Inspection of (6) shows that $v(x, t) \rightarrow 0$ as $t \rightarrow \infty$, and so $v(x, t)$ is called a **transient solution**. But observe in (8) that $u(x, t) \rightarrow \psi(x)$ as $t \rightarrow \infty$. In the context of solving forms of the heat equation, $\psi(x)$ is called a **steady-state solution**.

Time-Dependent PDE and BCs We turn now to a method for solving some kinds of BVPs that involve a *time-dependent nonhomogeneous equation* and *time-dependent boundary conditions*. A problem similar to (1),

$$\begin{aligned} k \frac{\partial^2 u}{\partial x^2} + F(x, t) &= \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0 \\ u(0, t) &= u_0(t), \quad u(L, t) = u_1(t), \quad t > 0 \\ u(x, 0) &= f(x), \quad 0 < x < L, \end{aligned} \quad (9)$$

describes the temperatures of a rod of length L but in this case the heat-source term F and the temperatures at the two ends of the rod can vary with time t . Intuitively one might expect that the line of attack for this problem would be a natural extension of the procedure that worked in Example 1, namely, seek a solution of the form $u(x, t) = v(x, t) + \psi(x, t)$. While this form of the solution is correct in some instances, it is usually not possible to find a function of two variables $\psi(x, t)$ that reduces a problem for $v(x, t)$ to a homogeneous one. To understand why this is so, let's see what happens when $u(x, t) = v(x, t) + \psi(x, t)$ is substituted into the PDE in (9). Because

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 \psi}{\partial x^2} \quad \text{and} \quad \frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} + \frac{\partial \psi}{\partial t}, \quad (10)$$

the BVP (9) becomes

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} + k \frac{\partial^2 \psi}{\partial x^2} + F(x, t) &= \frac{\partial v}{\partial t} + \frac{\partial \psi}{\partial t} \\ v(0, t) + \psi(0, t) &= u_0(t), \quad v(L, t) + \psi(L, t) = u_1(t) \\ v(x, 0) &= f(x) - \psi(x, 0). \end{aligned} \quad (11)$$

The boundary conditions on v in (11) will be homogeneous if we demand that

$$\psi(0, t) = u_0(t), \quad \psi(L, t) = u_1(t). \quad (12)$$

Were we, at this point, to follow the same steps in the method used in Example 1, we would try to force the problem in (11) to be homogeneous by requiring $k\psi_{xx} + F(x, t) = \psi_t$ and then imposing the conditions in (12) on the solution ψ . But in view of the fact that the defining equation for ψ is itself a nonhomogeneous PDE, this is an unrealistic expectation. So we try an entirely different tack by simply *constructing* a function ψ that satisfies both conditions given in (12). One such function is given by

$$\psi(x, t) = u_0(t) + \frac{x}{L}[u_1(t) - u_0(t)]. \quad (13)$$

Reinspection of (11) shows that we have gained some additional simplification with this choice of ψ because $\psi_{xx} = 0$. We now start over. This time if we substitute

$$u(x, t) = v(x, t) + u_0(t) + \frac{x}{L}[u_1(t) - u_0(t)] \quad (14)$$

the boundary-value problem (11) then becomes

$$\begin{aligned} k \frac{\partial^2 v}{\partial x^2} + G(x, t) &= \frac{\partial v}{\partial t}, \quad 0 < x < L, \quad t > 0 \\ v(0, t) &= 0, \quad v(L, t) = 0, \quad t > 0 \\ v(x, 0) &= f(x) - \psi(x, 0), \quad 0 < x < L, \end{aligned} \quad (15)$$

where $G(x, t) = F(x, t) - \psi_t$. While the problem (15) is still nonhomogeneous (the boundary conditions are homogeneous but the partial differential equation is nonhomogeneous) it is a problem that we can solve.

The Basic Strategy The solution method for (15) is a bit involved, so before illustrating with a specific example, we first outline the basic strategy:

Make the assumption that time-dependent coefficients $v_n(t)$ and $G_n(t)$ can be found such that both $v(x, t)$ and $G(x, t)$ in (15) can be expanded in the series

$$v(x, t) = \sum_{n=1}^{\infty} v_n(t) \sin \frac{n\pi}{L} x \quad \text{and} \quad G(x, t) = \sum_{n=1}^{\infty} G_n(t) \sin \frac{n\pi}{L} x, \quad (16)$$

where $\sin(n\pi x/L)$, $n = 1, 2, 3, \dots$, are the eigenfunctions of $X'' + \lambda X = 0$, $X(0) = 0$, $X(L) = 0$ corresponding to the eigenvalues $\lambda_n = \alpha_n^2 = n^2\pi^2/L^2$. This Sturm-Liouville problem would have been obtained had separation of variables been applied to the associated homogeneous BVP of (15). In (16), observe that the assumed series $v(x, t) = \sum_{n=1}^{\infty} v_n(t) \sin \frac{n\pi}{L} x$ already satisfies the boundary conditions in (15). Now substitute this series for $v(x, t)$ into the nonhomogeneous PDE in (15), collect terms, and equate the resulting series with the actual series expansion found for $G(x, t)$.

In the next example we illustrate this method by solving a special case of (9).

EXAMPLE 2 Time-Dependent Boundary Condition

Solve

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\ u(0, t) &= \cos t, \quad u(1, t) = 0, \quad t > 0 \\ u(x, 0) &= 0, \quad 0 < x < 1. \end{aligned}$$

SOLUTION We match this problem with (9) by identifying $k = 1$, $L = 1$, $F(x, t) = 0$, $u_0(t) = \cos t$, $u_1(t) = 0$, and $f(x) = 0$. We begin with the construction of ψ . From (13) we get

$$\psi(x, t) = \cos t + x[0 - \cos t] = (1 - x)\cos t,$$

and then, as indicated in (14), we use

$$u(x, t) = v(x, t) + (1 - x)\cos t \quad (17)$$

and substitute the quantities

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} + (1 - x)(-\sin t),$$

$$u(0, t) = v(0, t) + \cos t, \quad u(1, t) = v(1, t) \text{ and } u(x, 0) = v(x, 0) + 1 - x$$

into the given problem to obtain the BVP for $v(x, t)$:

$$\begin{aligned}\frac{\partial^2 v}{\partial x^2} + (1 - x)\sin t &= \frac{\partial v}{\partial t}, \quad 0 < x < 1, \quad t > 0 \\ v(0, t) &= 0, \quad v(1, t) = 0, \quad t > 0 \\ v(x, 0) &= x - 1, \quad 0 < x < 1.\end{aligned}\tag{18}$$

The eigenvalues and eigenfunctions of the Sturm-Liouville problem

$$X'' + \lambda X = 0, \quad X(0) = 0, \quad X(1) = 0$$

are found to be $\lambda_n = \alpha_n^2 = n^2\pi^2$ and $\sin n\pi x$, $n = 1, 2, 3, \dots$. With $G(x, t) = (1 - x)\sin t$ we assume from (18) that for fixed t , v and G can be written as Fourier sine series:

$$v(x, t) = \sum_{n=1}^{\infty} v_n(t) \sin n\pi x \tag{19}$$

$$\text{and} \quad (1 - x)\sin t = \sum_{n=1}^{\infty} G_n(t) \sin n\pi x. \tag{20}$$

By treating t as a parameter, the coefficients G_n in (20) can be computed:

$$G_n(t) = \frac{2}{1} \int_0^1 (1 - x)\sin t \sin n\pi x dx = 2\sin t \int_0^1 (1 - x)\sin n\pi x dx = \frac{2}{n\pi} \sin t.$$

$$\text{Hence,} \quad (1 - x)\sin t = \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin t \sin n\pi x. \tag{21}$$

We can determine the coefficients $v_n(t)$ by substituting (20) and (21) back into the PDE in (18). To that end, the partial derivatives of v are

$$\frac{\partial^2 v}{\partial x^2} = \sum_{n=1}^{\infty} v_n(t)(-n^2\pi^2)\sin n\pi x \quad \text{and} \quad \frac{\partial v}{\partial t} = \sum_{n=1}^{\infty} v'_n(t)\sin n\pi x. \tag{22}$$

Writing the PDE in (18) as $v_t - v_{xx} = (1 - x)\sin t$ and using (21) and (22) we get

$$\sum_{n=1}^{\infty} [v'_n(t) + n^2\pi^2 v_n(t)] \sin n\pi x = \sum_{n=1}^{\infty} \frac{2\sin t}{n\pi} \sin n\pi x.$$

We then equate the coefficients of $\sin n\pi x$ on each side of the equality to get the linear first-order ODE

$$v'_n(t) + n^2\pi^2 v_n(t) = \frac{2\sin t}{n\pi}.$$

Proceeding as in Section 2.3, we multiply the last equation by the integrating factor $e^{n^2\pi^2 t}$ and rewrite it as

$$\frac{d}{dt} [e^{n^2\pi^2 t} v_n(t)] = \frac{2}{n\pi} e^{n^2\pi^2 t} \sin t.$$

Integrating both sides, we find that the general solution of this equation is

$$v_n(t) = 2 \frac{n^2\pi^2 \sin t - \cos t}{n\pi(n^4\pi^4 + 1)} + C_n e^{-n^2\pi^2 t},$$

where C_n denotes the arbitrary constant. Therefore the assumed form of $v(x, t)$ in (19) can be written

$$v(x, t) = \sum_{n=1}^{\infty} \left[2 \frac{n^2\pi^2 \sin t - \cos t}{n\pi(n^4\pi^4 + 1)} + C_n e^{-n^2\pi^2 t} \right] \sin n\pi x. \tag{23}$$

The coefficients C_n can be found by applying the initial condition $v(x, 0)$ to (23). From the Fourier sine series,

$$x - 1 = \sum_{n=1}^{\infty} \left[\frac{-2}{n\pi(n^4\pi^4 + 1)} + C_n \right] \sin n\pi x \quad (24)$$

we see that the quantity in the brackets represents the Fourier sine coefficients b_n for $x - 1$. That is,

$$\frac{-2}{n\pi(n^4\pi^4 + 1)} + C_n = 2 \int_0^1 (x - 1) \sin n\pi x \, dx \quad \text{or} \quad \frac{-2}{n\pi(n^4\pi^4 + 1)} + C_n = \frac{-2}{n\pi}.$$

Therefore,
$$C_n = \frac{2}{n\pi(n^4\pi^4 + 1)} - \frac{2}{n\pi}.$$

By substituting the last result into (23) we obtain a solution of (18):

$$v(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \left[\frac{n^2\pi^2 \sin t - \cos t + e^{-n^2\pi^2 t}}{n(n^4\pi^4 + 1)} - \frac{e^{-n^2\pi^2 t}}{n} \right] \sin n\pi x.$$

At long last, then, it follows from (17) that the desired solution $u(x, t)$ is

$$u(x, t) = (1 - x) \cos t + \frac{2}{\pi} \sum_{n=1}^{\infty} \left[\frac{n^2\pi^2 \sin t - \cos t + e^{-n^2\pi^2 t}}{n(n^4\pi^4 + 1)} - \frac{e^{-n^2\pi^2 t}}{n} \right] \sin n\pi x. \quad \equiv$$

If the boundary-value problem has homogeneous boundary conditions and a time-dependent term $F(x, t)$ in the PDE, then there is no actual need to change the dependent variable through the substitution $u(x, t) = v(x, t) + \psi(x, t)$. For example, if both u_0 and u_1 are 0 in a problem such as (9), then it follows from (13) that $\psi(x, t) = 0$. The method of solution then begins by assuming an appropriate orthogonal series expansions for $u(x, t)$ and $F(x, t)$ as in (16), where the symbols v and G in (16) are naturally replaced by u and F , respectively.

EXAMPLE 3 Time-Dependent PDE and Homogeneous BCs

Solve
$$\frac{\partial^2 u}{\partial x^2} + (1 - x) \sin t = \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0$$

$$u(0, t) = 0, \quad u(1, t) = 0, \quad t > 0 \quad (25)$$

$$u(x, 0) = 0, \quad 0 < x < 1.$$

SOLUTION Except for the initial condition, the BVP (25) is basically (18). As pointed out in the paragraph preceding this example, because the boundary conditions are both homogeneous we have $\psi(x, t) = 0$. Thus all steps in Example 2 used in the solution of (18) are the same *except* the initial condition $u(x, 0) = 0$ indicates that the analogue of (24) is then

$$0 = \sum_{n=1}^{\infty} \left[\frac{-2}{n\pi(n^4\pi^4 + 1)} + C_n \right] \sin n\pi x.$$

We conclude from this identity that the coefficient of $\sin n\pi x$ must be 0 and so

$$C_n = \frac{2}{n\pi(n^4\pi^4 + 1)}.$$

Hence a solution of (25) is

$$u(x, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} \left[\frac{n^2\pi^2 \sin t - \cos t}{n(n^4\pi^4 + 1)} + \frac{e^{-n^2\pi^2 t}}{n(n^4\pi^4 + 1)} \right] \sin n\pi x. \quad \equiv$$

In Problems 13–16 in Exercises 13.6 you are asked to construct $\psi(x, t)$ as illustrated in Example 2. In Problems 17–20 of Exercises 13.6 the given boundary conditions are homogenous and so you can start as we did in Example 3 with the assumption that $\psi(x, t) = 0$.

REMARKS

Don't put any special emphasis on the fact that we used the heat equation throughout the foregoing discussion. The method outlined in Example 1 can be applied to the wave equation and Laplace's equation as well. See Problems 1–12 in Exercises 13.6. The method outlined in Example 2 is predicated on time dependence in the problem and so is not applicable to BVPs involving Laplace's equation.

13.6

Exercises

Answers to selected odd-numbered problems begin on page ANS-32.

In Problems 1–12, proceed as in Example 1 of this section to solve the given boundary-value problem.

In Problems 1 and 2, solve the heat equation $ku_{xx} = u_t$, $0 < x < 1$, $t > 0$ subject to the given conditions.

- $u(0, t) = 100$, $u(1, t) = 100$
 $u(x, 0) = 0$
- $u(0, t) = u_0$, $u(1, t) = 0$
 $u(x, 0) = f(x)$

In Problems 3 and 4, solve the heat equation (2) subject to the given conditions.

- $u(0, t) = u_0$, $u(1, t) = u_0$
 $u(x, 0) = 0$
- $u(0, t) = u_0$, $u(1, t) = u_1$
 $u(x, 0) = f(x)$
- Solve the boundary-value problem

$$k \frac{\partial^2 u}{\partial x^2} + Ae^{-\beta x} = \frac{\partial u}{\partial t}, \quad \beta > 0, 0 < x < 1, t > 0$$

$$u(0, t) = 0, \quad u(1, t) = 0, \quad t > 0$$

$$u(x, 0) = f(x), \quad 0 < x < 1,$$

where A is a constant. The PDE is a form of the heat equation when heat is generated within a thin rod due to radioactive decay of the material.

- Solve the boundary-value problem

$$k \frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}, \quad 0 < x < \pi, t > 0$$

$$u(0, t) = 0, \quad u(\pi, t) = u_0, \quad t > 0$$

$$u(x, 0) = 0, \quad 0 < x < \pi.$$

The PDE is a form of the heat equation when heat is lost by radiation from the lateral surface of a thin rod into a medium at temperature zero.

- Find a steady-state solution $\psi(x)$ of the boundary-value problem

$$k \frac{\partial^2 u}{\partial x^2} - h(u - u_0) = \frac{\partial u}{\partial t}, \quad 0 < x < 1, t > 0$$

$$u(0, t) = u_0, \quad u(1, t) = 0, \quad t > 0$$

$$u(x, 0) = f(x), \quad 0 < x < 1.$$

- Find a steady-state solution $\psi(x)$ if the rod in Problem 7 is semi-infinite extending in the positive x -direction, radiates from its lateral surface into a medium at temperature zero, and

$$u(0, t) = u_0, \quad \lim_{x \rightarrow \infty} u(x, t) = 0, \quad t > 0$$

$$u(x, 0) = f(x), \quad x > 0.$$

- When a vibrating string is subjected to an external vertical force that varies with the horizontal distance from the left end, the wave equation takes on the form

$$a^2 \frac{\partial^2 u}{\partial x^2} + Ax = \frac{\partial^2 u}{\partial t^2},$$

where A is constant. Solve this partial differential equation subject to

$$u(0, t) = 0, \quad u(1, t) = 0, \quad t > 0$$

$$u(x, 0) = 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \quad 0 < x < 1.$$

10. A string initially at rest on the x -axis is secured on the x -axis at $x = 0$ and $x = 1$. If the string is allowed to fall under its own weight for $t > 0$, the displacement $u(x, t)$ satisfies

$$a^2 \frac{\partial^2 u}{\partial x^2} - g = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < 1, \quad t > 0,$$

where g is the acceleration of gravity. Solve for $u(x, t)$.

11. Find the steady-state temperature $u(x, y)$ in the semi-infinite plate shown in **FIGURE 13.6.1**. Assume that the temperature is bounded as $x \rightarrow \infty$. [Hint: Use $u(x, y) = v(x, y) + \psi(y)$.]

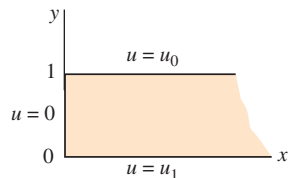


FIGURE 13.6.1 Semi-infinite plate in Problem 11

12. The partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -h,$$

where $h > 0$ is a constant, occurs in many problems involving electric potential and is known as **Poisson's equation**. Solve the above equation subject to the conditions

$$u(0, y) = 0, \quad u(\pi, y) = 1, \quad y > 0$$

$$u(x, 0) = 0, \quad 0 < x < \pi.$$

In Problems 13–16, proceed as in Example 2 to solve the given boundary-value problem.

13. $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0$

$$u(0, t) = \sin t, \quad u(1, t) = 0, \quad t > 0$$

$$u(x, 0) = 0, \quad 0 < x < 1$$

14. $\frac{\partial^2 u}{\partial x^2} + 2t + 3tx = \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0$

$$u(0, t) = t^2, \quad u(1, t) = 1, \quad t > 0$$

$$u(x, 0) = x^2, \quad 0 < x < 1$$

15. $\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < 1, \quad t > 0$

$$u(0, t) = 0, \quad u(1, t) = \sin t, \quad t > 0$$

$$u(x, 0) = 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \quad 0 < x < 1$$

16. $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0$

$$u(0, t) = 1 - e^{-t}, \quad u(1, t) = 1 - e^{-t}, \quad t > 0$$

$$u(x, 0) = 0, \quad 0 < x < 1$$

In Problems 17–20, proceed as in Example 3 to solve the given boundary-value problem.

17. $\frac{\partial^2 u}{\partial x^2} + xe^{-3t} = \frac{\partial u}{\partial t}, \quad 0 < x < \pi, \quad t > 0$

$$u(0, t) = 0, \quad u(\pi, t) = 0, \quad t > 0$$

$$u(x, 0) = 0, \quad 0 < x < \pi.$$

18. $\frac{\partial^2 u}{\partial x^2} + xe^{-3t} = \frac{\partial u}{\partial t}, \quad 0 < x < \pi, \quad t > 0$

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = 0, \quad \left. \frac{\partial u}{\partial x} \right|_{x=\pi} = 0, \quad t > 0$$

$$u(x, 0) = 0, \quad 0 < x < \pi$$

19. $\frac{\partial^2 u}{\partial x^2} - 1 + x - x \cos t = \frac{\partial u}{\partial t}, \quad 0 < x < 1, \quad t > 0$

$$u(0, t) = 0, \quad u(1, t) = 0, \quad t > 0$$

$$u(x, 0) = x(1 - x), \quad 0 < x < 1$$

20. $\frac{\partial^2 u}{\partial x^2} + \sin x \cos t = \frac{\partial^2 u}{\partial t^2}, \quad 0 < x < \pi, \quad t > 0$

$$u(0, t) = 0, \quad u(\pi, t) = 0, \quad t > 0$$

$$u(x, 0) = 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0, \quad 0 < x < \pi$$

13.7 Orthogonal Series Expansions

INTRODUCTION For certain types of boundary conditions, the method of separation of variables and the superposition principle lead to an expansion of a function in an infinite series that is *not* a Fourier series. To solve the problems in this section we shall utilize the concept of orthogonal series expansions or generalized Fourier series developed in Section 12.1.